

Storing Sensor Data *in* IoT Platforms Using Arduino

The Arduino board is a classic example of open source hardware that can be programmed and customised for use with various operating systems. It is commonly used to transmit IoT data to the cloud for analysis, storage and processing.



Open source software is very widely adopted throughout the world but open source hardware (OSH) is still undergoing research. Sometimes called free and open source hardware (FOSH), OSH refers to the firmware, chips, motherboard and sensors that are available for programming and customising, without any issues of proprietary licences and distribution. Open source hardware (comprising chips, sensors and boards, which can be programmed on demand to suit various needs) is commonly used in electronics and communications engineering.

Types of open source hardware

Different types of open source hardware associated with the development and programming of electronic boards are available for specific and on-demand applications, and can be deployed for integrated circuits on motherboards. Open source hardware can be used in the following fields.

- **Computers:** In desktop or laptop computing devices, much research is going on for the customisation of motherboards with real-time sensor devices so that live signals can be fetched in computers. These live signals may be related to satellite data, global positioning systems (GPS), temperature, humidity, environmental factors, ultrasonic signals, and so on.

- **Electronics:** This is the key domain in which electronic boards are customised using programming languages and scripts.
- **Mechatronics:** This is the merger of mechanical and electronics engineering.

Arduino

Arduino (<https://www.arduino.cc>) is an OSH and electronics platform for real-time applications.

It is available in many variants, and can be selected depending upon the type of application for which it is required. There are different boards available that can be programmed for specific applications.

Arduino boards are available for:

- Beginners
- Enhanced features
- Internet of Things (IoT)
- Education
- Wearable devices

Arduino UNO is one of the widely used microcontroller boards for beginners. This board can be easily programmed using the Arduino IDE, available at <https://www.arduino.cc/en/Main/Software>. The Arduino IDE software is available for multiple platforms including Windows, Mac OSX and Linux. C and C++ are



Figure 1: Logo of open source hardware by Open Source Hardware Association (OSHWA)